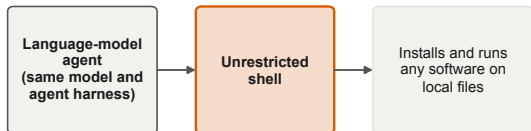


a

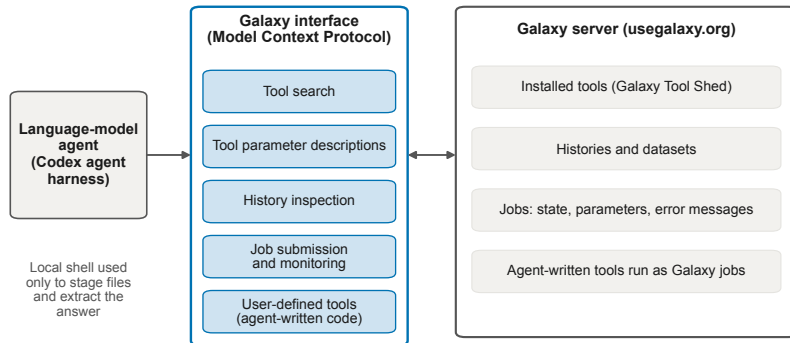
Open-ended code (reference condition)



Recorded: commands, scripts, command output and final files; no record of individual analysis jobs

b

Galaxy-mediated condition



Local shell used only to stage files and extract the answer

Recorded: tool identities and versions, requested and resolved parameters, datasets, job states and error messages

c Three benchmarks, five agent configurations, three replicates of every task

	BixBench-Verified-50		CompBioBench		IWC	
	Open-ended code	Galaxy	Open-ended code	Galaxy	Open-ended code	Galaxy
GPT-5.5	150	150	300	300	30	30
GPT-5.6 Sol	150	150	300	300	30	30
GPT-5.6 Luna	150	150	300	300	30	30
DeepSeek V4 Pro	150	150	300	300	30	30
DeepSeek V4 Pro, Claude Code harness (superseded; reported separately)	150	150	–	–	–	–
GPT-6 Astra (open-ended code only; not paired)	–	–	100	–	–	–

Numbers are archived runs: tasks × 3 replicates, per configuration and environment

BixBench-Verified-50

50 tasks; platform-neutral questions
Endpoint: answer accepted or rejected by the original evaluator

CompBioBench

100 tasks; platform-neutral questions
Endpoint: aggregate score (answers credited, of 100)

IWC (Intergalactic Workflow Commission)

10 tasks derived from published Galaxy workflows
Endpoint: agreement with the workflow's reference output (0 to 1)

Archive: 4,240 runs · 4,228 agent traces
23,080 Galaxy analysis jobs · 69,812 Galaxy interface calls